

NEYAZ AHMED

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Education

Master of Science in Applied Mathematics and Computational Science

North South University, Dhaka, Bangladesh

- **GPA:** 3.56 (ongoing)
- **Thesis** (*in progress*): Comparative Analysis of Phase Change Material (PCM) and Nano-Enhanced Phase Change Material (NEPCM) Thermal Storage Performance in Cavities with Serrated Heat Sources: Effects of Geometric Morphology and Material Properties
Methods: transient melting simulations in ANSYS Fluent using the enthalpy–porosity formulation
- **Advisor:** Dr. Preetom Nag

Bachelor of Science in Petroleum and Mining Engineering

Shahjalal University of Science & Technology (SUST), Sylhet, Bangladesh

- **Thesis:** Consumer Gas Risk Analysis in Bangladesh: A Case Study of Utility Gas and Liquefied Petroleum Gas (LPG) Safety
- **Advisor:** Dr. Labiba Nusrat Jahan

Standardized Test Scores	
GRE	IELTS Academic
Total: 329 Quantitative: 170 (91st percentile) Verbal: 159 (80th percentile)	Overall: 8.0 Listening: 8.0 Reading: 9.0 Writing: 6.5 Speaking: 7.5

Research Experience

Graduate Research Assistant

Department of Mathematics & Physics, North South University

Finite-volume CFD and data-driven surrogate modeling of convective heat transfer in cavities, porous media, and geothermal systems. Manuscripts under review:

- *Data-Driven Surrogate Modeling of Cylinder-Wise Heat Transfer in Conjugate MHD Mixed Convection of a Hybrid Ferrofluid*
- *Numerical Investigation of a Double-Tube Vertical Geothermal Exchanger in Pile Foundations*
- *Numerical Investigation of Natural Convection in a Porous Square Cavity with Discrete Cylindrical Heat Sources*

Machine Learning — Big Matrix Lab

[GitHub: Repositories](#)

- CFD investigation of temperature and velocity profiles in an AGR reactor: impact of rib geometry and temperature-profile optimization via the Kriging method
- Statistical benchmarking of machine learning models for building energy-efficiency prediction, with 10-fold cross-validation
- Comparative analysis of machine learning models for breast cancer diagnosis: performance benchmarking and statistical validation

Applied Mathematics Project

North South University

- Comparative analysis of fluid flow through a 3D 90° angular piping bend using numerical simulation

Research Interests

Computational Fluid Dynamics · Convective Heat Transfer · Phase Change & Thermal Energy Storage · Magnetohydrodynamics · Lattice Boltzmann Methods · Scientific Machine Learning & Surrogate Modeling

Technical Skills

CFD & Multiphysics:	ANSYS Fluent, COMSOL Multiphysics, Lattice Boltzmann Method (in-house D2Q9 solver, C)
High-Performance Computing:	OpenMP, MPI, CUDA
Programming Languages:	Python (NumPy, SciPy, PyTorch, TensorFlow), MATLAB/Simulink, C
Scientific Computing:	Numerical Methods, Finite Volume & Finite Difference Methods
Machine Learning for Physical Systems:	Hybrid physics-ML modeling, surrogate modeling
Tools & Collaboration:	Git (GitHub), $\text{\LaTeX}$

Teaching & Mentorship Experience

- **Teaching Assistant**, Applied Mathematics, North South University (2025 – Present)
Supported graduate-level courses, lab sessions, and problem-solving tutorials.
- **One-on-One Mentoring** in Mathematics and Programming (2021 – Present)
Provided individual guidance and project supervision for students.

Professional Experience

Engineering Intern, Baset Prokoushol Limited, Dhaka, Bangladesh
Conducted site investigations, soil testing, and field data collection using surveying equipment.

Honors & Awards

- **Champion**, Hult Prize at SUST — advanced to Regional Finals, Mumbai, India
- **National Board Scholarship**, Higher Secondary Certificate Examination (2017)
- **Bronze Medal**, Duke of Edinburgh International Award Programme (2017)
- **National Board Scholarship**, Secondary School Certificate Examination (2015)
- **Champion**, National Mathematics Olympiad (2013)

Professional Training

Technical Institute for Chemical Industries (TICI), Bangladesh
Industrial water treatment; process safety & accident prevention; PLC/SCADA automation; industrial pumps; electrical maintenance

Professional Development & Memberships

Courses: Machine Learning (Stanford University / Coursera, 2020); CS50's Introduction to Programming with Python (Harvard University / edX)

Memberships: Society of Petroleum Engineers (SPE), 2017 – Present

References

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